Tech Trend Project

Cryptocurrency as an Asset Class

Risk–Return Profile, Diversification, and Regulation of an Emerging Alternative

Finance Analysis

Asset Class Evaluation

Cryptocurrency as an Asset Class: Investor Perspective

Blockchain-Based Digital Asset

Cryptocurrency is secured by cryptography and operates on decentralized blockchain networks, creating a new category of digital value storage.

Diversification Benefits

Portfolio diversification potential exists but varies significantly across market regimes, with correlations rising during financial stress.

Bitcoin as Market Proxy

Bitcoin dominates liquidity and market capitalization (~56%), serving as the primary benchmark for crypto market performance.

Institutional Adoption

Regulatory frameworks and institutional infrastructure increasingly shape crypto investability, custody solutions, and risk management.

High-Volatility Alternative

Crypto exhibits unique return drivers distinct from traditional assets, with volatility profiles characteristic of alternative investments.

Key Insight

"The question is no longer whether crypto is 'real,' but whether it behaves like a distinct investable bucket with a repeatable role in portfolio construction."

— Academic Research, Review of Financial Studies

Market Scale & Access

Market Size and Institutional Adoption

$2.4T
Global Crypto Market Cap
BTC Dominance: 56%

$126T
Global Equity Markets
~52× larger than crypto

$145T
Global Fixed Income
~60× larger than crypto

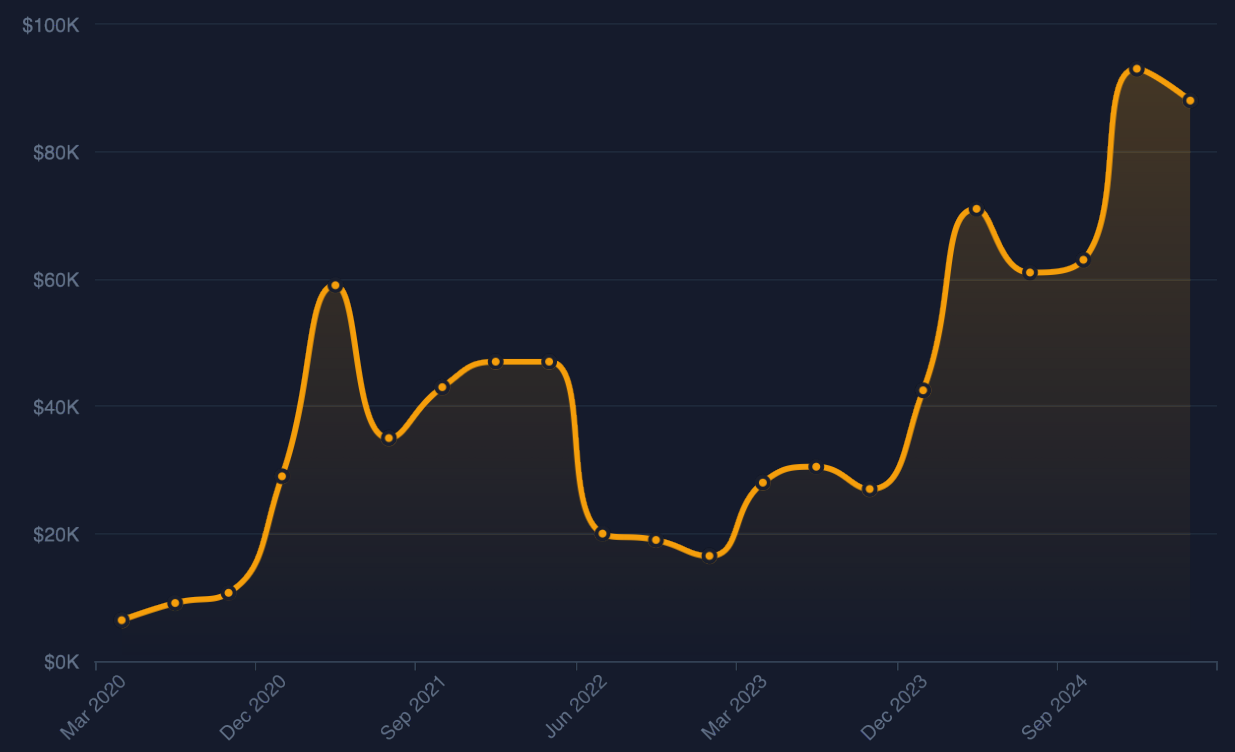
Jan 2024
U.S. Spot Bitcoin
ETF Approval

Institutional Access Evolution

- SEC approval of spot Bitcoin ETPs in January 2024 marked a regulatory milestone
- Major financial institutions now offer crypto custody and trading services
- Increased institutional participation improves market liquidity and price discovery

Portfolio Positioning
Crypto is typically treated as a **"satellite allocation"** in diversified portfolios due to its size relative to traditional markets.

Bitcoin Price Performance (5-Year)



5-Year Return:
~1,275%

Cross-Asset Analysis

Risk–Return Characteristics Compared to Traditional Assets

5-Year Comparative Performance (2020–2025)

Asset	Market Size	5-Yr Return	Volatility	Corr. to S&P 500
Bitcoin	$1.3T	~50-60%	~60-80%	0.3-0.5
S&P 500	$50T+	~14-18%	~15-20%	1.00
Bonds (Agg)	$145T	~0-3%	~4-6%	-0.1-0.2
Gold	$14T+	~6-10%	~12-15%	0.0-0.2

* Volatility measured as annualized standard deviation of daily returns

Key Risk Observations

- **Extreme Volatility:** Crypto volatility is 3-4× higher than equities
- **Stress Correlation:** Correlation with equities rises during market turbulence
- **Regime Dependence:** Diversification benefits weaken when most needed

Portfolio Implications

1-5%

Optimal Allocation Range

- Position sizing critical for risk management
- Regular rebalancing recommended
- Governance framework essential

IMF Finding: Crypto-equity spillovers have increased over time and intensify during turbulent periods.

Strategic Assessment

Opportunities, Risks, and Ethical Considerations

Positive Impacts

- **Financial Innovation:** New market infrastructure and settlement technologies
- **Cross-Border Efficiency:** Reduced friction in international payments
- **Financial Inclusion:** Access for unbanked populations globally
- **Institutional Opportunities:** New asset class for portfolio diversification

Key Risks

- **Extreme Volatility:** Price swings can exceed 50% in short periods
- **Regulatory Uncertainty:** Evolving global policy landscape
- **Cybersecurity & Fraud:** Exchange hacks and scam risks persist
- **Environmental Impact:** Mining consumes 0.6-2.3% of U.S. electricity
- **Market Integrity:** DeFi vulnerabilities and governance concerns

Regulatory Landscape

US **SEC Bitcoin ETF Approval**
January 2024 — Increased institutional access while emphasizing risks

EU **MiCA Regulation**
Effective December 2024 — Comprehensive crypto asset framework

BC **Basel Capital Rules**
Effective January 2025 — Bank prudential standards for crypto exposures

Conclusion

Cryptocurrency has evolved into a measurable alternative asset class with high growth potential and distinct risk-return characteristics.

However, its portfolio role must be carefully managed due to **extreme volatility**, **regulatory uncertainty**, and **evolving market structure**. Prudent allocation (1-5%) with robust governance frameworks is essential for institutional adoption.